
Expansion Impact Analysis – Touch N Go Third Location

**Forecasting Cannibalization,
Complementary Growth, and Customer
Attendance at a Proposed Laguna Niguel
Location**

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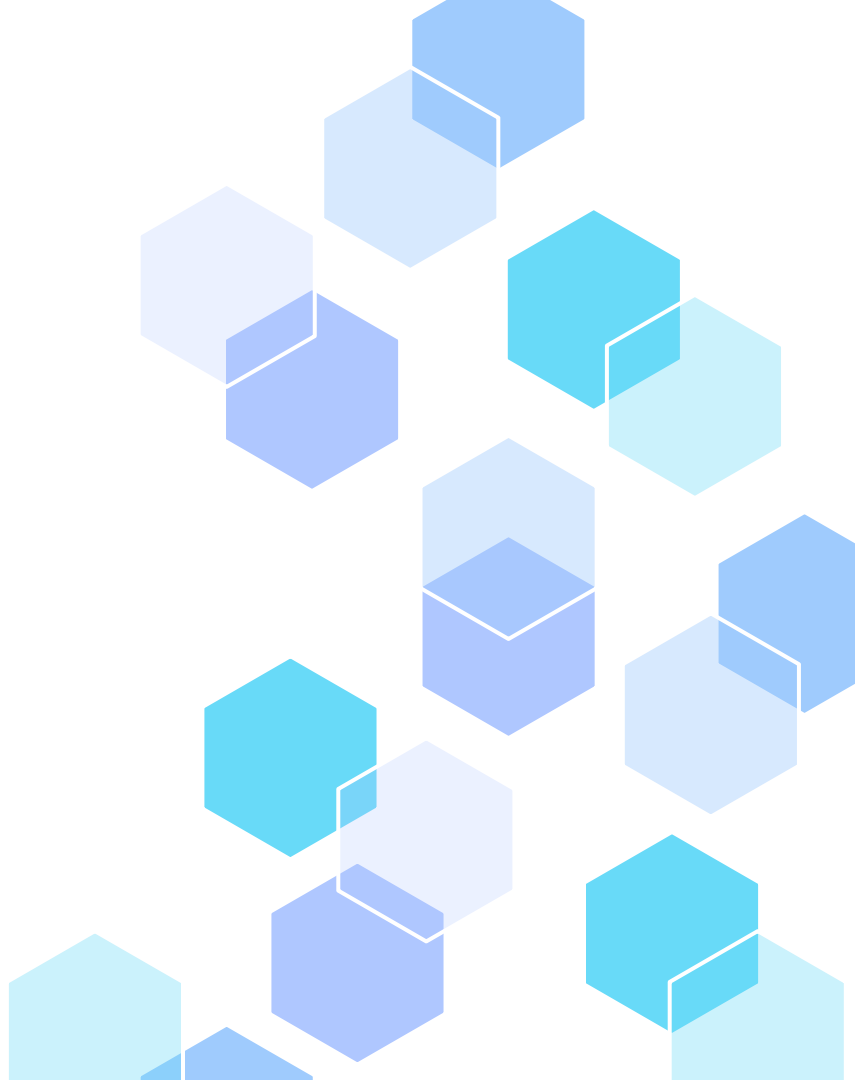
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01 Project Introduction

Business background, the problem, and project
goals



Introduction



What is Touch N Go Soccer?

- Youth soccer training organization operating in Southern California
- Two current locations: Corona, CA and Tustin, CA
- Plans to expand to a third location

What does Touch N Go Soccer offer?

- Youth soccer training sessions across multiple session types
- Membership plans for recurring customers
- Coached training programs for various age groups
- Youth soccer leagues
- Youth academy program in partnership with local school districts, where students attend school for half the day and train at Touch N Go Soccer for the other half

Why does this matter?

- Management needs data-driven insights to evaluate expansion
- Key concerns are cannibalization, complementary growth, and total expected attendance at a new location

The Problem

Cannibalization

- Will a Laguna Niguel location pull existing customers away from Corona or Tustin?

Complementary Growth

- Will the new location attract customers who currently don't attend due to distance?

No Established Baseline

- No models exist specifically for youth sports training facilities
- This Project adapts existing frameworks to fill that gap



Project Goals

Goal 1

Current Location Analysis:
Understand attendance patterns, session popularity, and where customers come from at Corona and Tustin

Goal 2

Customer Behavior & Retention:
Identify the factors that drive customers to return, churn, and convert to memberships

Goal 3

Expansion Forecasting:
Use insights from existing locations to estimate cannibalization risk, identify complementary growth potential, and evaluate the impact of a Laguna Niguel expansion



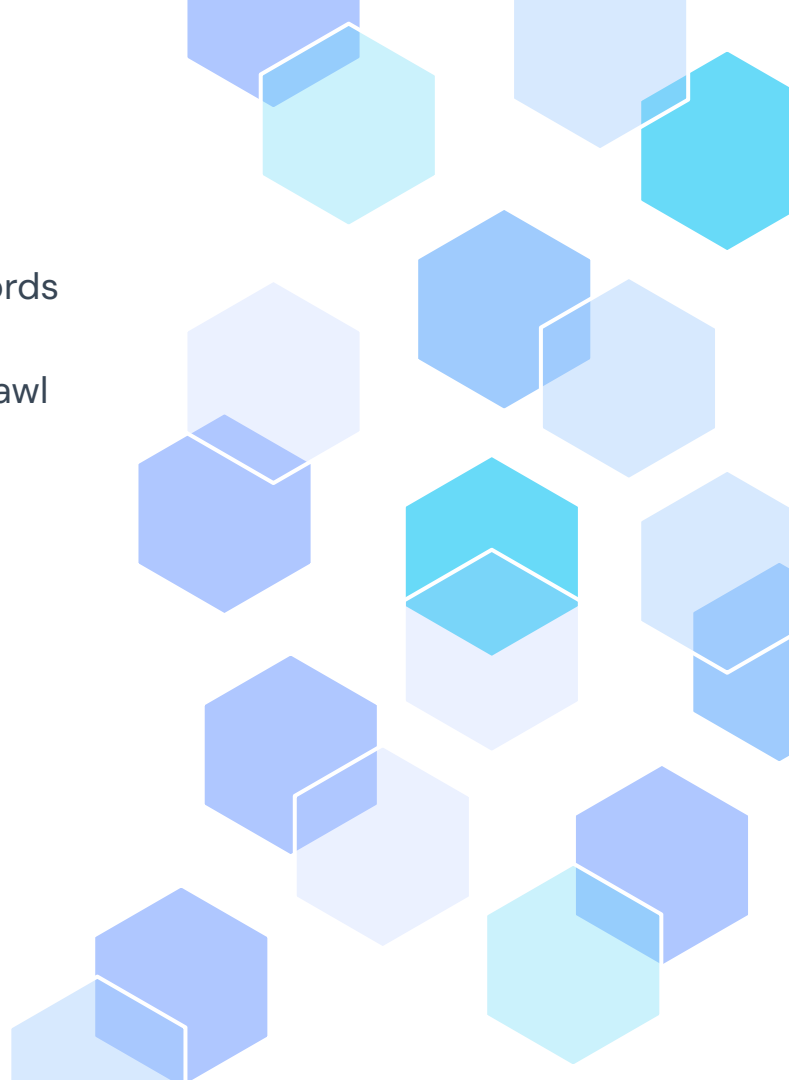
02 Data Collection

Data Sources, Collection Process,
and Dataset Construction

Data Sources Overview

Internal (Mindbody)

- Customer profiles, full visit history, and membership records for Corona and Tustin
- Data collected through a custom web scraper built to crawl the Mindbody platform
- Supplemented by manual exports through Mindbody's built-in reporting tools



The Mindbody Web Scraper

Why a Scraper Was Needed:

- Mindbody is a third-party platform with limited bulk export capabilities
- The Mindbody API requires a partner program signup that was not accessible
- A custom scraper was built to collect all necessary visit data directly from the platform



The Mindbody Web Scraper

How It Worked

Selenium is a browser automation tool built into python. It lets you write code that controls a real web browser, clicking buttons, filling out forms, navigating pages exactly like a human would, but automatically.

Selenium connected to an already open and authenticated Chrome session using a remote debugging port, avoiding login detection entirely.

The scraper read a CSV file containing every client ID exported from Mindbody. Using each ID, it constructed the direct URL to that client's visits page, requiring no manual navigation.

For every client, Selenium navigated directly to their visits page using the client ID in the URL, located the Show All Dates button and Export button using their unique HTML identifiers, and triggered a CSV download. Files were renamed using the client ID and saved locally.

To handle thousands of clients, multiple workers ran simultaneously, each assigned a separate chunk of the client list. A progress file tracked completed clients so the scraper could resume if interrupted.

The Mindbody Web Scraper

The Process Step by Step

- Step 1: Log into Mindbody portal
- Step 2: Load the full client list and iterate through each client
- Step 3: For each client, open their account page
- Step 4: Navigate to the visits section and scrape all visit records
- Step 5: Save each visit to a local database with customer ID, date, session type, and location
- Step 6: Move to the next client and repeat

Result

Corona and Tustin visit histories fully collected

Corona: 18,017 visits

Tustin: 114,799 visits

Total raw: 132,816 visits

Data Sources Overview

External

- **Census Data:** Youth population ages 5–17, median household income, households with children, and population density by ZIP code (U.S. Census Bureau's American Community Survey (ACS) 5-Year API)
- **School Counts:** Number of elementary, middle, and high schools by ZIP code sourced (National Center for Education Statistics)
- **Drive Times:** Calculated from every ZIP code to Corona, Tustin, and Laguna Niguel (OSRM, a free open-source routing tool built on OpenStreetMap road data)



Three Datasets Created

ZIP Code Level

443 rows, one row per ZIP code. Contains demographics, drive times, and aggregated customer demand. Used for the Huff Model and expansion forecasting.

Customer Level

7,607 rows, one row per customer. Contains behavioral features, membership status, and churn labels. Used for retention and membership conversion modeling.

Visit Level

126,321 rows, one row per visit. Contains session type, coach, date, and return behavior. Used for attendance trends and session analysis.

Dataset 1: ZIP Code Level

443 rows, one row per ZIP code across Southern California (After Cleaning)

How It Was Built:

- ZIP codes were extracted from the Corona and Tustin client mailing lists
- Filtered to the Southern California range
- Matched against the U.S. Census Bureau ACS API
- Of 557 valid SoCal ZIP codes found, 443 returned usable Census data.
- The remaining 114 were PO boxes or non-residential ZIP codes with no demographic records available.

Dataset 1: ZIP Code Level

External — Census & Schools

Youth population ages 5–17
Median household income
Total households
Percent households with children
Elementary schools
Middle schools
High schools
Total schools

External — Drive Times

Drive time to Corona (min)
Drive time to Tustin (min)
Drive time to Laguna Niguel (min)

Aggregated — Combined

Total customers
Total visits
Customers per 1,000 youth

Aggregated — Per Location

Total customers
Customers with visits
Total visits
Avg visits per customer
Active members
Churned customers
Retention rate
Membership conversion rate
Applied separately for Corona and Tustin

Dataset 2: Customer Level

7,607 rows, one row per customer combining Corona and Tustin records (After Cleaning)

How it was built:

- Each customer from both locations was pulled from the Mindbody mailing lists and merged with their visit history and membership records using their client ID.
- From the visit history, behavioral features were calculated per customer (visits in the first 30 days, days to second visit, unique session types, unique coaches, and average visit frequency)
- From the membership records, each customer was flagged as active member, past member, or non-member. (Customers with a membership but zero visits were removed as they were likely staff accounts.)
- Drive time to their attended facility was added by matching each customer's ZIP code to the distances dataset.
- Two target variables were created: a churn flag for any customer with no visit in the last 90 days, and a membership conversion flag for any customer who ever purchased a membership.

Dataset 2: Customer Level

Behavioral Features

Total visits

Days active

Days since last visit

Average visits per month

Visits in first 30 days

Days to second visit

Visit streak

Months since first visit

Unique session types attended

Unique coaches seen

First session type

First month of enrollment

Geographic & Profile

Drive time to attended facility

Facility (Corona or Tustin)

Target Variables

Churned — no visit in last 90 days

Converted to membership — yes or no

Cleaning Notes

444 customers missing ZIP code — kept for behavioral modeling

Zero-visit accounts removed

Duplicate IDs across locations kept — different people

Dataset 3: Visit Level

126,321 rows, one row per visit across both Corona and Tustin (After Cleaning)

How it was built:

- The raw visit files scraped from Mindbody were cleaned and combined into a single visit level dataset.
- Each row represents one individual visit by one customer.
- Session descriptions were categorized into standardized session types to allow consistent analysis across both locations.
- A returned within 30 days flag was engineered for each visit by checking whether that customer came back within 30 days of that specific session.
- This captures short term retention at the individual visit level rather than the customer level

Dataset 3: Visit Level

Visit Identity

Client ID
Location
Facility (Corona or Tustin)
Status
Pricing option
Description
Session category

Time Features

Date
Day of week
Time
Hour
AM or PM
Month
Month name
Year

Coach & Retention

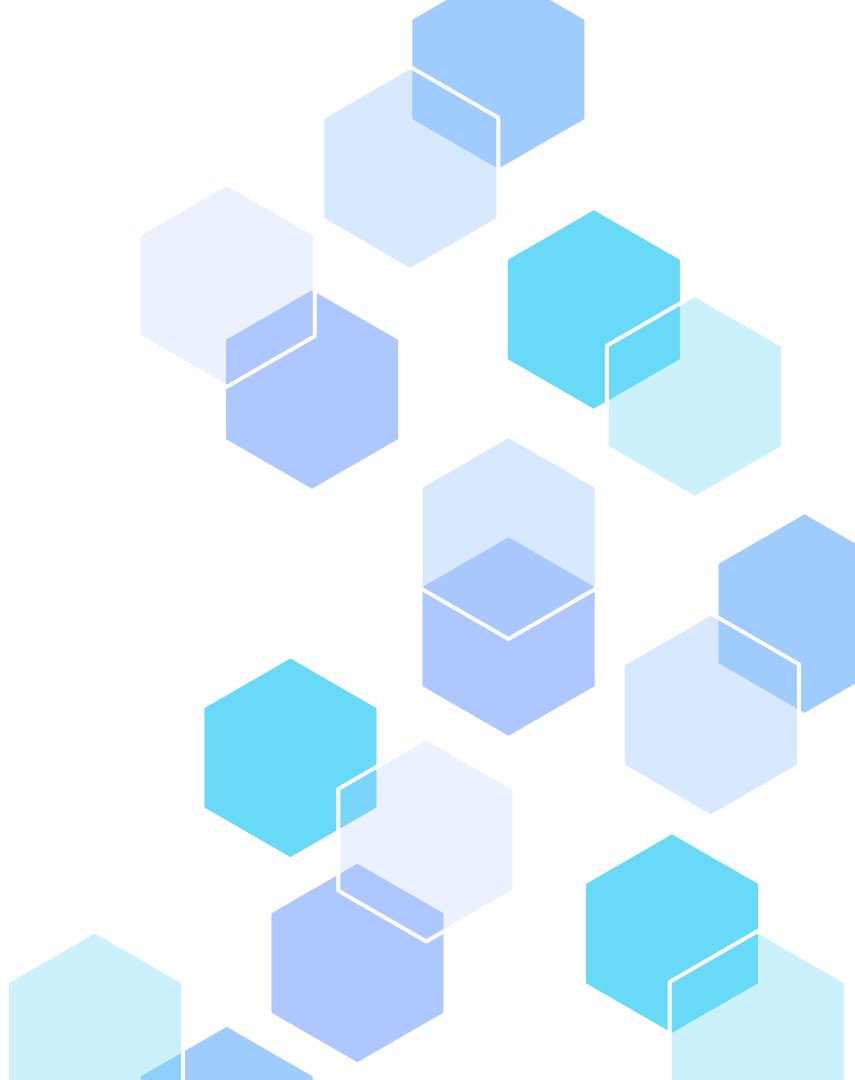
Coach
Returned within 30 days

Cleaning Notes

Non-training sessions removed
Session descriptions standardized
132,816 raw visits reduced to 126,321
created_by column dropped

03 Models and Methodology

Model selection, methodology, and the math
behind each approach



Why Three Different Models

Huff Model

Spatial choice probability

Given where a customer lives, which location will they choose? Estimates the probability each ZIP code gravitates toward Corona, Tustin, or Laguna Niguel based on distance and attractiveness.

Regression

Interpretable demand forecasting

Which variables most strongly predict attendance, churn, and membership? Identifies coefficients for each demographic and behavioral feature so findings can be explained clearly to management.

Random Forest

Non-linear pattern detection

Are there complex patterns in the data that regression misses? Captures non-linear relationships between demographics and customer behavior, and ranks feature importance independently.

Using all three creates a layered analysis

If multiple models agree on a finding, the finding is stronger

The Huff Model

- A spatial interaction model originally developed for retail trade area analysis (Huff, 1963)
- Estimates the probability that a customer from a given ZIP code will choose a specific location over competing alternatives
- Applied to all three locations simultaneously: Corona, Tustin, and Laguna Niguel
- ZIPs where Laguna has the highest probability represent new market territory. ZIPs where Tustin or Corona currently dominate but shift toward Laguna represent cannibalization risk
- Limitation: Since all Touch N Go locations offer the same services, attractiveness was set equally to 1 for all locations. The model therefore functions purely as a distance-based spatial choice model. The sole factor determining which location a customer chooses is drive time.

THE FORMULA

$$P_i = \frac{A_i}{D_i^\lambda}$$

$$\sum (A_i / D_i^\lambda)$$

Applied across all ZIP codes simultaneously for Corona, Tustin, and Laguna Niguel

P_i

Probability

The likelihood that a customer from a given ZIP code will choose location i over all competing alternatives. The output is a probability between 0 and 1 for each location, and all probabilities sum to 1.

A_i

Attractiveness

Set equal to 1 for all three locations. Since Corona, Tustin, and Laguna Niguel all offer the same services and programs, no location has an inherent advantage over another. Setting attractiveness to 1 removes it as a variable entirely, making drive time the sole factor determining which location a customer chooses.

D_i

Drive time

The drive time in minutes from the customer's ZIP code centroid to location i , calculated using OSRM. Drive time is the strongest predictor of customer choice and is used in place of straight-line distance to better reflect real-world travel behavior.

λ

Distance decay

Controls how sensitive customers are to distance. A higher lambda means customers strongly prefer the closest location and are less willing to travel further. A lower lambda means distance matters less and attractiveness plays a bigger role in the decision.

Regression Models

THE FORMULA

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n$$

Y

Outcome to predict

β

Learned coefficients

X

Input features

β_0

Baseline value

Regression is a statistical method that learns from historical data to predict an outcome based on a set of input variables.

How Coefficients Are Determined

- The model is trained on historical customer data and searches for the combination of coefficient values that produces predictions closest to what actually happened.
- It does this by measuring the error between each prediction and the actual outcome, then repeatedly adjusting the coefficients in the direction that reduces that error.
- This process continues until the error is minimized and the model has learned the relationship between each feature and the outcome.

Regression Models

THE FORMULA

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n$$

Y

Outcome to predict

β

Learned coefficients

X

Input features

β_0

Baseline value

How to Read the Coefficients

- Positive coefficient: that feature increases the predicted outcome
- Negative coefficient: that feature decreases the predicted outcome
- Example: Customers who visit more in their first 30 days have a negative coefficient for churn, meaning early frequent visits are strongly associated with staying long term.

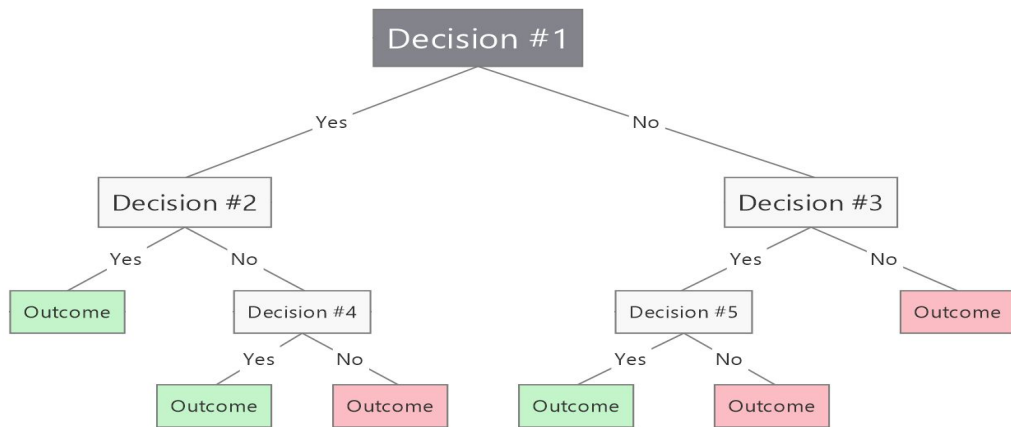
Models Used

- Linear Regression — predicts a continuous outcome, used here to predict average visits per month
- Logistic Regression — predicts a yes or no outcome, used here to predict churn and membership conversion
- Strength: Fully interpretable and directly explainable to management
- Weakness: Assumes a straight line relationship between features, missing more complex relationships

Decision Tree Logic

At each level the tree asks the following: Which piece of information about a customer would help us predict whether they churn most quickly?

It tests every feature and picks the one that does the best job of separating churned customers from retained customers. That feature becomes the question at that branch. Then it repeats the process on each resulting group, going deeper until the groups are clean enough to make a confident prediction.



***Note: scikit-learn's Random Forest uses Gini impurity by default, which measures the same concept as entropy using a simpler formula.

Decision Tree Logic

ENTROPY

$$-(p_1 \times \log_2 p_1 + p_2 \times \log_2 p_2)$$

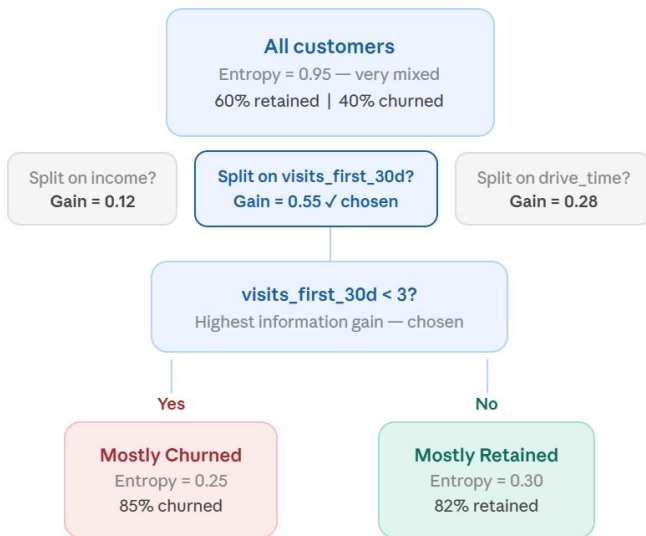
0 = perfectly pure

1 = completely mixed

INFORMATION GAIN

$$\text{Entropy}(\text{parent}) - \text{avg Entropy}(\text{children})$$

The feature with the highest gain is chosen at each level



***Note: scikit-learn's Random Forest uses Gini impurity by default, which measures the same concept as entropy using a simpler formula.

Why Random Forest

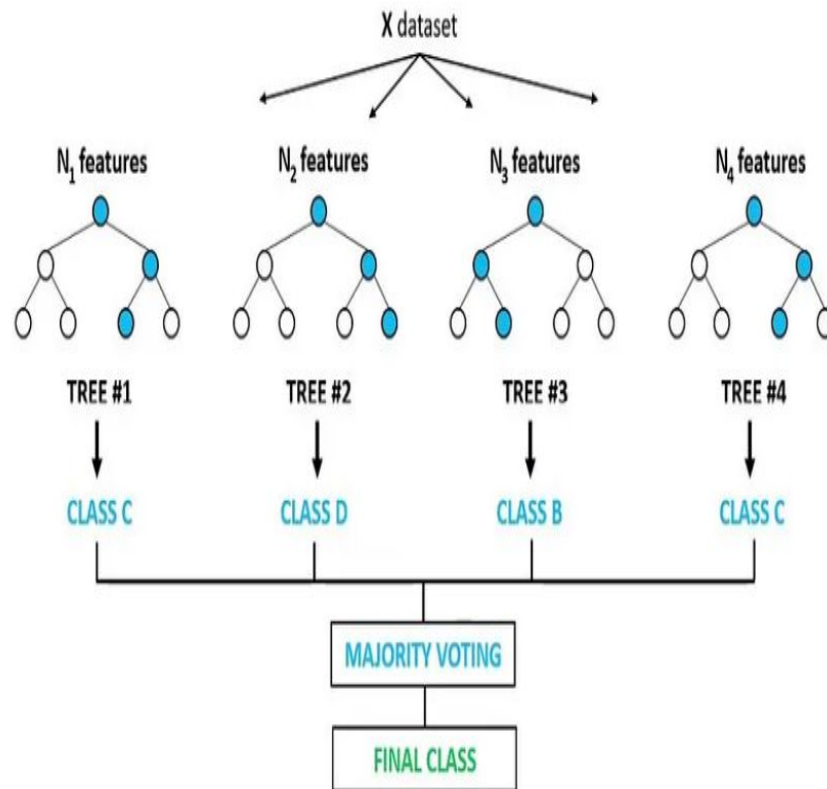
Random Forest builds hundreds of decision trees, each trained on a different random subset of the data. Every tree makes its own prediction and the final answer is a majority vote across all trees, making it more reliable than any single tree alone

A single tree memorizes the training data too closely and performs poorly on new data. Building hundreds of trees on different random samples and averaging their predictions fixes this and removes potential overfitting

Customer behavior is complex and nonlinear. A customer's likelihood of churning is not a straight line relationship with any single feature. Random Forest captures these complex interactions automatically, something regression cannot do

Strength: Handles nonlinear patterns and provides feature importance rankings

Weakness: Less interpretable than regression and requires more data to train effectively



Analysis and Visualizations for Management

All results are available on an interactive dashboard hosted publicly for management access.

https://davcarbajal32.github.io/Touchngo-Expansion-Impact-Analysis/analysis/goal1/goal1_outputs/heatmaps_dashboard.html



04 Goal 1:

Current Location

Analysis

session popularity, attendance
trends, geographic heatmaps

Goal 1 Takeaways and Analysis (Session Analysis)

66.7%

Corona — Youngers
of all visits

58.5%

Tustin — Youngers
of all visits

3x

Youngers growth
since 2022 at Tustin

HIGHEST RETURN RATES

Ball Mastery (Int/Young)	91.1%
Speed of Play (Adv/Young)	91.0%
Shooting (Int/Young)	90.5%
Camp (outlier)	61.8%

FIRST SESSION VS TOTAL VISITS

Ball Mastery (Int/Young)	31.1 avg visits
Speed of Play (Adv/Young)	29.6 avg visits
First Touch (Beg/Young)	27.0 avg visits
Shooting (Beg/Old)	9.8 avg visits

LOWEST CHURN RATE BY FIRST SESSION

Juniors	69.5% churn
Ball Mastery (Beg/Young)	70.8% churn
Passing (Int/Young)	73.2% churn
Strength & Conditioning	96.5% churn

MEMBERSHIP CONVERSION BY FIRST SESSION

Shooting (Beg/Old)	5.1%
Shooting (Int/Young)	4.3%
Juniors	3.1%
Overall avg conversion	~2.0%

Goal 1 Takeaways and Analysis (Attendance Trends)

2,961

Tustin peak visits
July 2024

1,459

Corona peak visits
Jan 2026 — 16 months old

85.8%

30-day return rate
Stable for 4+ years

March

Strongest month
9,167 visits since 2022

TUSTIN GROWTH STORY

Opened Oct 2018 with ~100 visits/month
Near-zero in April 2020 — COVID shutdown
Recovered and grew steadily through 2021-2022
Explosive growth in 2023-2024, tripling volume
Peak: 2,961 visits in July 2024

CORONA RAMP TRAJECTORY

Opened Oct 2024 with 10 visits
642 visits by November — immediate demand
Crossed 1,000 visits by January 2025
Summer 2025 plateau around 1,100-1,200
New peak: 1,459 in January 2026

30-DAY RETURN RATE STABILITY

Tustin has held 84-90% consistently since 2022
Corona started at 70% in Nov 2024 and quickly caught up
Both locations now tracking at the same level
Strong signal of consistent product quality

LONG-TERM COHORT RETENTION

2021-2022 joiners by 2025 **42-47% still active**
2023 joiners by 2025 **39% still active**
2019 joiners by 2025 **~5% still active**
Long-term re-engagement is the key challenge

Goal 1 Takeaways and Analysis (Customer Behavior)

20 min

Median drive time
Both locations

256 days

Tustin median lifetime
103 days at Corona

91.4%

July churn rate
Highest at both locations

48.7%

Feb churn rate
Best month at Corona

VISIT FREQUENCY DISTRIBUTION

1-2 visits (Tustin)	1,521 customers
3-5 visits (Tustin)	1,453 customers
21-50 visits (Tustin)	689 customers
200+ visits (Tustin)	80 customers

The largest group visits only 1-5 times — converting these early customers is the biggest opportunity

CUSTOMER LIFETIME

Corona median	103 days
Corona mean	161 days
Tustin median	256 days
Tustin mean	427 days

Mean far exceeds median — a small number of very long-term customers pull the average up significantly

DRIVE TIME FINDINGS

Both locations have nearly identical drive time profiles — median 18-20 minutes

Almost all customers live within 50 minutes of their facility

Very few outliers travel more than 60 minutes

Directly validates the Huff Model assumption — proximity is everything

CHURN BY ENROLLMENT MONTH

July enrollees (Corona)	86.2% churn
July enrollees (Tustin)	91.4% churn
Feb enrollees (Corona)	48.7% churn
Mar enrollees (Tustin)	72.6% churn

Summer enrollees are likely camp or trial customers who never intended to stay — winter and spring joiners are the committed base

Goal 1 Takeaways and Analysis (Coaches & Schedule)

24,035

Candice Silva
Tustin visits led

5,038

David Carbajal
Corona visits led

4PM

Peak training hour
Both locations

Monday

Busiest day
Both locations

TOP COACHES BY 30-DAY RETURN RATE (CURRENT COACHES ONLY)

% of this coach's sessions where the customer returned to the facility within 30 days

Paco Martinez (Tustin)	88.5%
Candice Silva (Tustin)	88.2%
Chris Ramirez (Tustin)	87.7%
Mario Luna (Tustin)	87.2%
David Carbajal (Tustin)	87.0%

TNG Staff (unassigned) 43-47%

COACH CONCENTRATION RISK

Top 3 coaches at Tustin — Candice Silva, Mario Luna, Brandon Segura — account for over 63,000 visits combined.

That is 58% of all Tustin visit volume concentrated in 3 coaches.

High dependency on key staff is a business risk at Laguna.

PEAK SCHEDULING WINDOW

Monday through Wednesday, 4PM to 6PM is the absolute hot zone at both locations.

4PM alone accounts for 6,042 visits at Corona and 35,130 at Tustin.

This after-school window is the core of the business — must be protected and prioritized at any new location.

WEEKEND OPPORTUNITY

Saturday morning 10-11AM shows meaningful activity at both locations.

Sunday is the quietest day but still generates 1,225 visits at Corona and 11,206 at Tustin.

Weekend morning slots are underutilized relative to demand — an expansion opportunity at Laguna.



05 Goal 2

Deeper Customer Behavior and Retention Analysis

What drives churn, visit
frequency, and membership
conversion

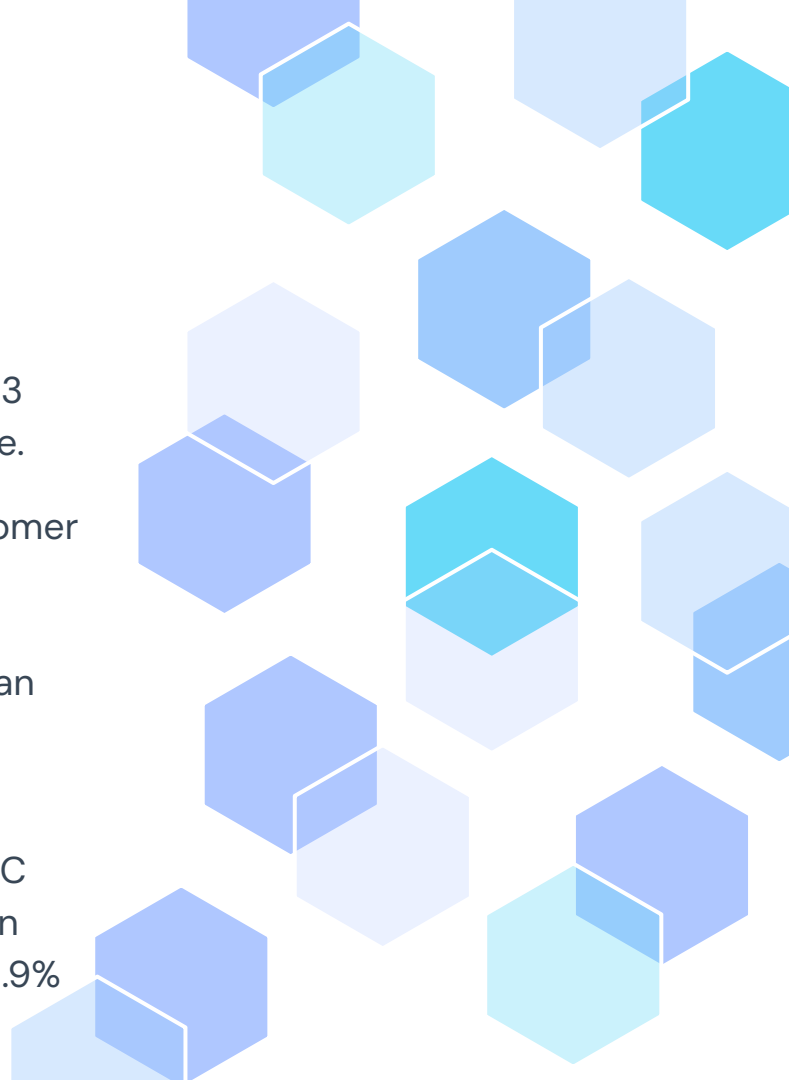
Model Overview

Three models were built for Goal 2.

Model A (Dataset 2)— Churn Prediction: Will this customer become inactive? Random Forest ROC-AUC 0.764 combined, 0.805 Corona, 0.767 Tustin. Logistic Regression ROC-AUC 0.703 combined. Trained on 6,086 customers with a 74.9% churn rate.

Model B (Dataset 2)— Membership Conversion: Will this customer purchase a membership? Random Forest ROC-AUC 0.909 combined, 0.947 Corona, 0.935 Tustin. Logistic Regression ROC-AUC 0.905 combined. Trained on 6,086 customers with an 11.6% membership rate.

Model C (Dataset 3)— Post-Visit Return: After a visit, will this customer come back within 30 days? Random Forest ROC-AUC 0.813 combined, 0.815 Corona, 0.805 Tustin. Logistic Regression ROC-AUC 0.788 combined. Trained on 92,973 visits with an 85.9% return rate.



Churn Prediction

Model Performance

- **Approach:** Random Forest (RF) vs. Logistic Regression (LR)
- **Result:** RF consistently outperformed LR (Combined ROC-AUC: 0.764 vs. 0.703)
- **Takeaway:** Nonlinear patterns exist; RF is the superior predictive tool

Feature Selection (All three models used only features knowable within a customer's first 30 days to prevent data leakage)

- **Dropped:** *total_visits: churned customers have fewer visits by definition, directly encoding the outcome. *days_since_last_visit tells the model whether someone already churned. *avg_visits_per_month and *months_since_first_visit both derived from full visit history.
- **Kept:** first_month, has_second_visit, visits_first_30d, drive_time, return signals within 7 and 30 days, first session category, first coach, facility, and demographic ZIP features.



Churn Prediction

Top Predictors (RF Feature Importance)

- **Enrollment Month (17.4%):** Summer enrollees are highest risk.
- **Second Visit (17.2%):** The "Make-or-Break" moment for retention.
- **First 30 Days (9.2%):** Early frequency correlates with long-term loyalty.
- **Drive Time (6.9%):** Distance significantly increases churn risk (esp. at Tustin).

Management Takeaway

- **The Second Visit is Everything:** `has_second_visit` is the strongest protective factor (-0.79).
- **Potential action:** Deploy follow-up incentives within 7 days of the initial visit to secure the second stay.



Membership Prediction

Model Performance

- **Accuracy:** Exceptionally high predictive power (RF ROC-AUC 0.909, LR ROC-AUC: 0.905).
- **Consistency:** RF and LR results align closely, indicating a clear conversion signal.
- **Takeaway:** Model identifies high-potential members with extreme confidence.

Feature Selection (All three models used only features knowable within a customer's first 30 days to prevent data leakage.)

- **Dropped:** *total_visits: churned customers have fewer visits by definition, directly encoding the outcome. *days_since_last_visit tells the model whether someone already churned. *avg_visits_per_month and *months_since_first_visit both derived from full visit history.
- **Kept:** first_month, has_second_visit, visits_first_30d, drive_time, return signals within 7 and 30 days, first session category, first coach, facility, and demographic ZIP features.



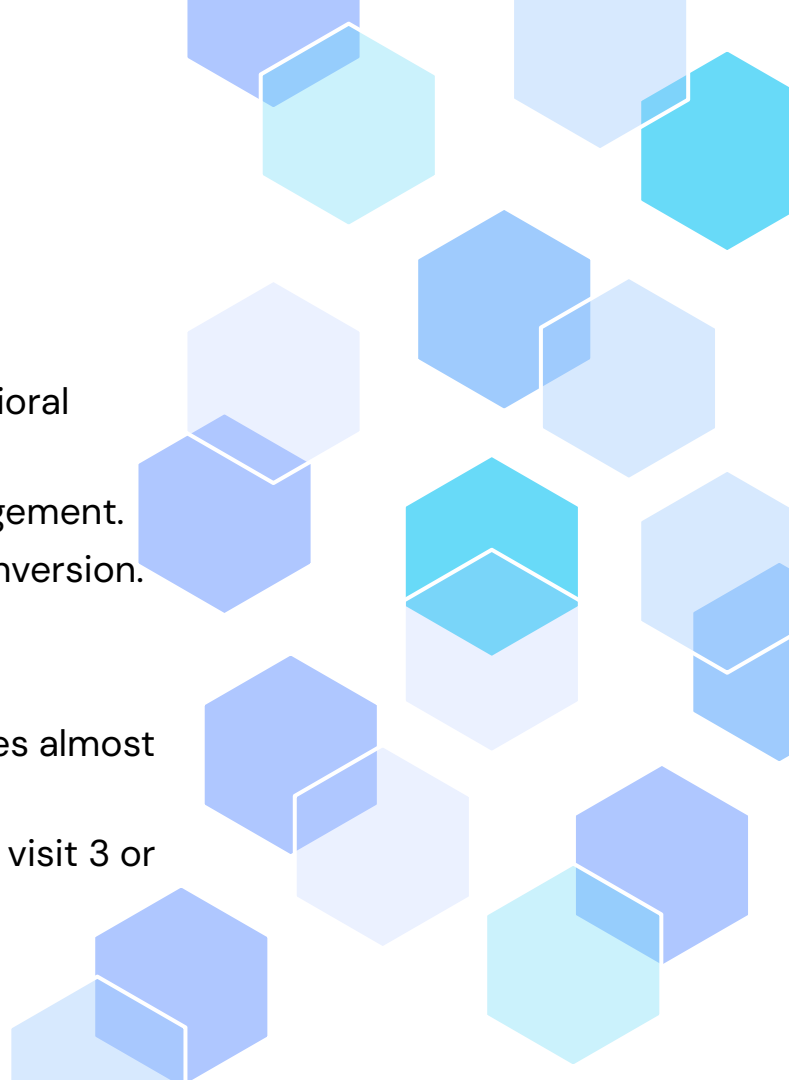
Membership Prediction

Top Predictors (RF Feature Importance)

- **Visits First 30 Days (37.9%):** The dominant engine of membership.
- **7-Day Return (15.1%):** Speed of return is a critical behavioral indicator.
- **Second Visit (8.4%):** Essential binary signal for re-engagement.
- **Drive Time (5.7%):** Proximity correlates strongly with conversion.

Management Takeaway

- **The Power of Frequency:** 5+ visits in month one indicates almost certain conversion.
- **Action:** Initiate structured membership conversations at visit 3 or 4 to strike while momentum is high.



Post-Visit Return Prediction

Model Performance

- **Accuracy:** Strong predictive power (RF ROC-AUC 0.813, LR ROC-AUC 0.788).
- **Consistency:** Consistent results across both locations; the same visit-level signals drive return behavior everywhere.
- **Takeaway:** The model effectively identifies loyal patterns, though first-visit returns remain the hardest to forecast.

Feature Selection (To prevent data leakage, the model excludes future visit data and only uses information available the moment a customer walks in)

- **Dropped:** am_pm (redundant with hour) and any metrics derived from visits that haven't happened yet.
- **Kept:** visit_rank, days_since_first_visit, is_first_visit, session category, coach, drive_time, hour, day of week, and month.



Post-Visit Return Prediction

Top Predictors (RF Feature Importance)

- **Visit Rank (40.1%):** The dominant predictor. Veteran customers almost always return; new customers are the primary source of uncertainty.
- **Tenure (20.4%):** `days_since_first_visit` confirms that the longer a customer stays, the more stable their return habit becomes.
- **Drive Time (8.4%):** Distance remains a persistent friction point even for established customers.
- **Camp Category (5.1%):** Camp attendees are significantly less likely to return than those in training sessions.

Management Takeaway

- **The First-Visit Risk:** ~15% of newcomers never return. Because this drop-off is driven by "invisible" factors (satisfaction, engagement), the first experience must be perfected.
- **The "Second Visit" Bridge:** Offer an immediate discount the next session. This incentivizes a second visit, allowing customers to experience different coaches and classes before deciding.

Action:

- **Front Desk:** Execute a high-touch handoff by physically walking to the floor to introduce new clients to their coaches, ensuring they feel welcomed rather than lost.
- **Coaches:** Implement mandatory "Pre-and-Post" scripts. Coaches should set goals before the session and provide direct feedback to the parent and child afterward to build an immediate personal connection.
- **Optimization:** Strategically assign the most engaging coaches and specific session types to first-timers to stabilize the high-risk 15% who currently drop off after a single visit.



06 Goal 3: Expansion Forecasting

Cannibalization vs
complementary growth, Laguna
demand forecast, and priority ZIP
codes

Model Overview

Dataset 1: All demographic and drive time features used for training. Huff probabilities from Model D output fed in as weights for Model E at prediction time.

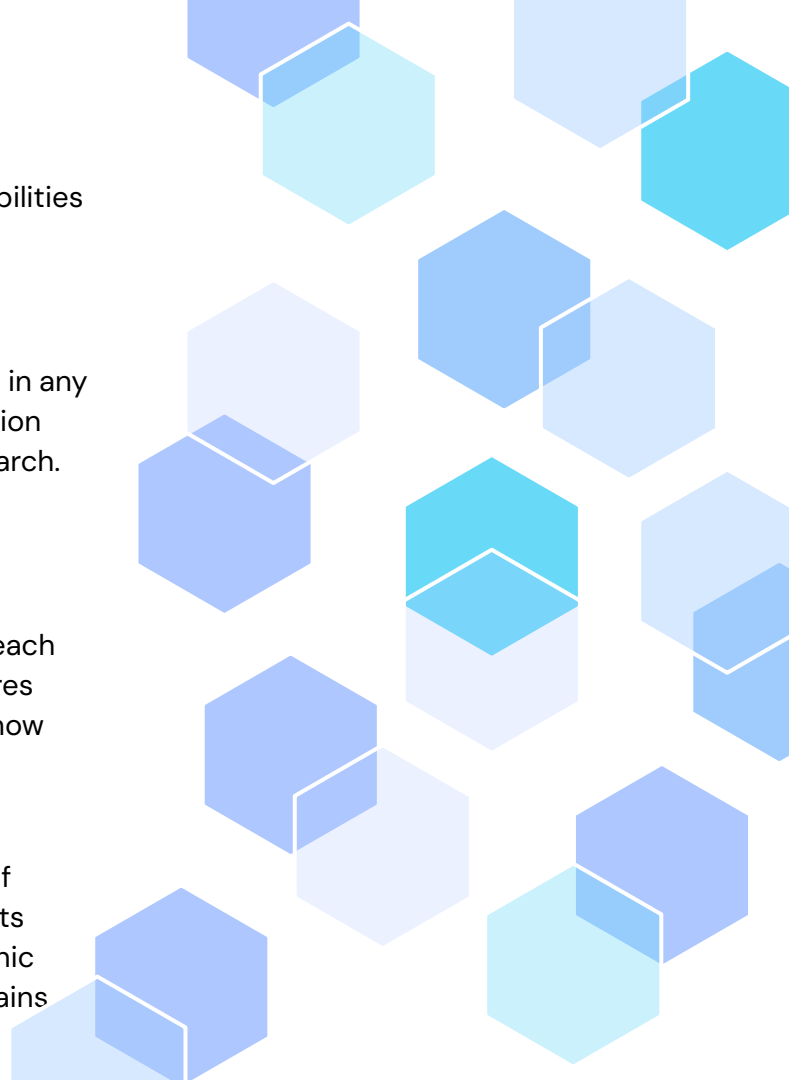
Model D — Huff Spatial Choice Model

Formula-based spatial interaction model estimating the probability a customer in any ZIP code would choose Corona, Tustin, or Laguna based on drive time and location attractiveness. A validated spatial economics formula widely used in retail research. Attractiveness set to 1 for all three locations.

Model E — ZIP-Level Demand Forecast

Random Forest Regressor trained on all 443 Southern California ZIP codes. For each ZIP, the model learned the relationship between demographic and spatial features such as drive time, youth population, household income, total households and how many Touch N Go customers that ZIP produced.

Once trained, it was applied to Laguna-area ZIPs to predict demand if Laguna opened. Those predictions were then multiplied by each ZIP's Huff probability of choosing Laguna, so a ZIP with a 40% Huff probability only contributes 40% of its predicted demand. This accounts for both demographic potential and geographic competition simultaneously. Cross-validated R^2 of 0.770 means the model explains 77% of the variance in customer demand across ZIP codes.



Huff Model: Where Would Customers Choose

ZIPs analyzed

443

Cannibalization (midpoint)

3,215 visits

Sensitivity range

2,630–3,664

TOP ZIPS BY P(LAGUNA)

92677 Laguna Niguel	99.4%
92629 Dana Point	85.9%
92624 Capistrano Beach	65.8%
92656 Aliso Viejo	65.4%
92675 San Juan Capistrano	60.0%

CANNIBALIZATION BREAKDOWN

Corona visits lost	324
Tustin visits lost	2,891
Total cannibalized	3,215

Note: 324 of Corona's losses come from a single borderline ZIP — 92530 Lake Elsinore — where Laguna is only 8 minutes farther than Corona.

ZIP codes throughout Laguna Niguel, Aliso Viejo, Mission Viejo, and Dana Point show Laguna probability near 85-99%. These areas are currently underserved by both existing locations.

Note: High P(Laguna) reflects geographic preference only. ZIPs with small youth populations (e.g., 92624 Capistrano Beach, youth pop 944) appear here but rank lower in Slide 46's priority list because total demand is volume-driven, not probability-driven.

Laguna Demand Forecast

Year 1 customers

791

Year 1 visits

13,229

Avg visits/customer

16.7

Model R²

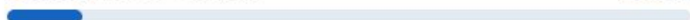
0.770

RF FEATURE IMPORTANCE

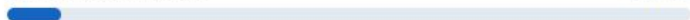
Min drive time to existing 68.7%



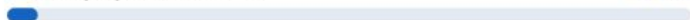
Drive time to Corona 10.8%



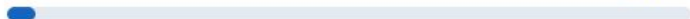
Total households 7.9%



Youth population 5-17 4.4%



Median household income 4.0%



FORECAST BREAKDOWN

Total customers 791

Cannibalized customers 232

From Corona 12

From Tustin 220

Net new customers 559

Net new visits 9,433

Drive time is the single most powerful predictor at 68.7% — directly validating the Huff model assumption and Goal 2 behavioral findings. Avg 16.7 visits per customer is consistent with Tustin's historical average.

Priority Zip Codes for Laguna Marketing

Top 5 ZIPs — customers

249

Top 5 ZIPs — visits

4,143

Share of total demand

31%

ZIP / City	Drive time	P(Laguna)	Customers	Visits	Priority
92677 Laguna N.	1.4 min	99%	85	1,420	HIGH
92656 Aliso Viejo	9.2 min	65%	66	1,096	HIGH
92629 Dana Point	8.1 min	86%	41	679	HIGH
92692 Mission Viejo	15.7 min	44%	29	476	MED
92653 Laguna Hills	10.1 min	56%	28	472	MED
92675 San Juan C.	13.6 min	60%	27	446	MED
92691 Mission Viejo	16.0 min	39%	24	400	MED
92694 Ladera Ranch	20.0 min	52%	20	327	MED
92688 RSM	17.9 min	40%	19	321	MED
92651 Laguna Beach	13.1 min	43%	18	298	MED

First marketing targets: soccer club outreach, school partnerships, and digital advertising in HIGH priority ZIPs before opening.

Cannibalization VS Complementary Growth

CANNIBALIZATION — 29%

3,796

visits redistributed from existing locations

From Tustin

3,576

From Corona

220

COMPLEMENTARY GROWTH — 71%

9,433

genuinely new visits to Touch N Go

Net new customers

559

Total Laguna visits

13,229

VISUAL SPLIT — YEAR 1 LAGUNA VISITS

71% net new

29% cannibalized

9,433 new visits

3,796 redistributed visits

Sensitivity validated: cannibalization ranges from 2,630 to 3,664 across beta values 1.0 to 3.0 — a tight range confirming the directional finding. Tustin bears nearly all risk. Corona is largely unaffected.

Cannibalization estimated at 3,215 visits by the Huff spatial baseline model and 3,796 visits by the Random Forest demand model. The agreement between two independent methodologies strengthens confidence in the directional finding

Largest Takeaways

EXPANSION IS JUSTIFIED

791 customers and 13,229 visits projected in Year 1. 71% of that demand is genuinely new to Touch N Go — the expansion is predominantly complementary, not cannibalistic.

CANNIBALIZATION IS MANAGEABLE

Tustin bears nearly all risk at 3,576 visits — only 3.3% of its current annual volume. Corona is largely unaffected at 220 visits due to its Inland Empire geography.

PROXIMITY DRIVES EVERYTHING

Drive time accounts for 68.7% of the Random Forest model's decisions — directly validating the Huff model and every Goal 2 behavioral finding. Distance is the business.

TARGET MARKET IS CLEAR

ZIPs 92677, 92656, 92629, 92692, and 92653 account for 31% of projected Year 1 demand. These are the first marketing, outreach, and school partnership targets.

LAUNCH STRATEGY MATTERS AS MUCH AS LOCATION

Based on Goal 2 findings, the first 30 days of a new Laguna customer's experience directly determine long-term retention and membership conversion. A structured onboarding program — follow-up after first visit, incentive to return within the first week, early membership offer — should be built before the doors open.

Conclusion

The Data Science Life Cycle

- **End-to-End Execution:** Completed the full real-world life cycle, including stakeholder requirements gathering, custom ETL (Extract, Transform, Load) pipeline development, model validation, and executive delivery.
- **Bridging Theory & Practice:** Translated classroom concepts to real-world application
- **Real-World Constraints:** Navigated the challenges of fragmented industry data (Mindbody) to build a clean, centralized database, moving beyond the "clean" datasets typically found in academic settings.

Strategic Impact

- **Expansion Viability:** Confirmed Laguna Niguel as a high-potential site with minimal cannibalization of the Tustin hub.
- **Predictive Accuracy:** Achieved a Cross-validated R^2 of 0.77, providing management with a high-confidence forecast for site demand.
- **Operational Insights:** Identified the "Second Visit" as the critical driver for retention, offering a specific target for front-desk operations.

Final Summary

- **Scalable Framework:** Delivered a modular analysis toolkit that can be potentially reused for future locations, demonstrating how data science drives efficient, low-risk business growth.
- **Professional Readiness:** Proven ability to connect technical methodology with actionable leadership strategy, preparing for professional application in data analytics.

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Thank You

Any questions?

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https://davcarbajal32.github.io/Touchngo-Expansion-Impact-Analysis/analysis/goal1/goal1_outputs/heatmaps_dashboard.html

<https://github.com/davcarbajal32/Touchngo-Expansion-Impact-Analysis.git>

